Proof of Concept: Automating Pesticide Label Data Extraction for xarvio Field Manager (xFM) Master Data

Preface: This document outlines a proof of concept for automating pesticide label data extraction for xarvio Field Manager. This initiative originated as part of an internal hackathon at BASF, where it was recognized for its potential value and subsequently selected as a project to pursue further. While I was involved in the initial hackathon and conceptualization of this proof of concept, other project responsibilities prevented my direct participation in its later stages of development. This document reflects the initial vision and proposed approach developed during the hackathon.

1. Introduction

Within the xarvio Field Manager (xFM) ecosystem, accurate and accessible input data are crucial for effective crop models and recommendation engines. A significant bottleneck exists in the management of this data: the manual effort required to ingest pesticide label information into our master data database. Pesticide labels provide the essential information and legal requirements for the application of these products. These labels contain the information on correct timing, rates, and restrictions that need to be incorporated into our recommendation engine to ensure that xFM provides recommendations to our customers. However, the information on pesticide labels that needs to be ingested into the database and used by our models is typically contained within tables in PDF documents and typically do not exist in a relational database. This unstructured format poses a significant challenge for integrating this data into the master data framework.

This report outlines the problems that the master data team has faced with manual extraction of pesticide labels and proposes an automated solution as a proof of concept and discusses the potential business value and impact of implementing this into the master data framework.

2. The Challenge: Unstructured Pesticide Label Data in xFM

The effectiveness of xFM recommendations powered by our crop, disease and weather models is linked to the specific requirements of the product labels (timing, re-application restrictions, amount of active ingredients, etc). However, to go beyond basic timing recommendations, that is, to link recommendations to specific products requires a complete and comprehensive database of label requirements.

The current reliance on unstructured pesticide label data presents two key challenges for xFM:

- **Manual Effort:** Integrating pesticide label information into the master data database is a labor-intensive process. For each region, domain experts must painstakingly read and interpret each label, manually transferring the information into a structured format. This process is time-consuming and costly, significantly hindering our ability to quickly onboard new products or update existing ones available in xFM.

    To provide a clearer picture of the manual effort involved, consider the typical workflow of the domain expert (note these are not data entry specialists):

1) **Label Sourcing**: The specialist gathers pesticide labels relevant to a specific crop and region. In the US, this typically involves a manual search of the EPA's website (https://ordspub.epa.gov/ords/pesticides/f?p=PPLS:1::::::).

2) **Label Acquisition**: Once identified, the domain expert downloads or obtains the physical copies of the pesticide labels, which are often in PDF format.

3) **Information Location**: The domain expert opens each label and manually locates the relevant information. This can be challenging due to the variability in label formats. They must navigate through dense text, tables, and diagrams to find details on:

    - Product name and registration number
    - Active ingredients and concentrations
    - Application rates and methods
    - Timing and frequency of application
    - Pre-harvest intervals (PHIs)
    - Restricted entry intervals (REIs)
    - Crop-specific instructions
    - Geographic restrictions

4) **Information Interpretation**: The specialist interprets the information, which may involve:

    - Decoding abbreviations and acronyms (e.g., "ai," "fl oz," "gpa")
    - Understanding complex phrases and conditional statements (e.g., "Repeat applications on 14-day to 28-day intervals as needed. Use the shorter interval and/or the higher rate when disease pressure is high")

5) **Data Entry**: The specialist manually enters the extracted information into a structured database. This requires them to:

    - Map the extracted information to the correct database fields
    - Adhere to specific data formats and validation rules
    - Ensure accuracy and consistency across all entries

6) **Quality Control**: The specialist reviews their entries to identify and correct any errors or omissions. This may involve comparing the database entries to the original labels and making necessary revisions.

Because domain experts are not trained data entry specialists, on average, it can take several hours for them to process a single label.

- **Scalability Issues:** xFM aims to be product agnostic. Considering the vast number of pesticide products used in a given region, the total time and cost associated with manual data entry become substantial.

3. Proposed Solution: Automated Label Data Extraction for xFM (Proof of Concept)

To address these challenges for xFM, a proof of concept for an automated pesticide label data extraction pipeline is presented. This pipeline leverages a combination of techniques to convert unstructured label data into a structured format that can be seamlessly integrated into master data:

- **3.1 Document Pre-processing:**
    - **PDF Processing and Table Extraction:** While pesticide labels must meet regulatory requirements, they lack a standardized digital format and are typically provided in PDF with varying layouts. Therefore the pipeline can begin by extracting both text and tables. For text extraction, standard libraries can be used. Tools like pdfplumber can be employed to extract tables, recognizing that many labels contain crucial information within tables.
        - **Table Relevance Assessment:** A key challenge is determining which extracted tables contain the relevant information. This could be addressed using a combination of techniques:
            - **Heuristic-based filtering:** Rules could be defined to filter tables based on column headers (e.g., "Crop," "Application Rate," "fl ozs/Acre") or keywords.
            - **Machine learning-based classification:** It is possible to train a model to classify tables as relevant or irrelevant based on their features (e.g. keywords, numeric values).
- **3.2 Information Extraction:**
    - **Advanced Table Extraction and Interpretation:** Beyond simply extracting table structures, the system must be able to interpret the meaning of table rows and columns. This is particularly important when information is spread across multiple rows or uses complex formatting (e.g. merged cells or spanned cells).
        - **For example consider the second row in Figure 1**
        - To correctly interpret this, the system needs to:
            - Identify information types (Application Directions, Use Restrictions).
            - Extract core Application Directions (timing, conditions for rate/interval).
            - Recognize individual Use Restrictions (minimum interval, maximum rates, etc.).
            - Associate values with units and active ingredients.

        This involves techniques like:

            - Parsing
            - Semantic analysis
            - Regular expressions
            - Contextual understanding

<table>
<tr><td>Corn
(all types)</td><td>Anthracnose
Colletotrichum graminicola
Gray leaf spot
Cercospora zeae-maydis
Northern corn leaf blight
Exserohilum turcicum
Northern corn leaf spot
Cochliobolus carbonum
Southern corn leaf blight
Bipolaris maydis
Tar Spot
Phyllachora maydis
Rust, common
Puccinia sorghi
Rust, southern
Puccinia polysora</td><td>4.5 to 6.5</td><td>2</td><td>13</td><td>21</td></tr>
<tr><td colspan="6">Application Directions: For optimal disease control, begin applications of Revylok prior to disease development if conditions are conducive for disease development.

Spray Interval
- Sweet corn: Apply at 7-day to 14-day intervals.
- Field corn, Popcorn, Silage, Seed corn: Apply at 14-day intervals.

Adjuvant or Crop Oil Use Limitation on Corn. Adjuvant crop damage can occur when an adjuvant or crop oil is used after the V8 stage and before the VT stage (the VT stage is defined as when the tassel's last branch is completely visible outside the whorl across the entire field). If an adjuvant or crop oil is used after the V8 stage and before the VT stage, the grower and the user are responsible for contacting the adjuvant source (adjuvant distributor, retailer, or manufacturer) for advice and confirmation that the adjuvant has been tested and proven to be safe for application from the V8 to VT corn stages. Refer to adjuvant and/or crop oil labels for specific use directions and restrictions. Always follow the most restrictive label.

Another fungicide or an insecticide labeled for use on corn may be included in the tank mix if needed. Refer to the product label for tank mix pesticide for specific use directions and restrictions. Always follow the most restrictive label.

Use Restrictions
• The minimum retreatment interval is 14 days for field corn and 7 days for sweet corn.
• Corn forage, fodder and stover MUST NOT be fed to livestock sooner than 21 days after last application.
• DO NOT apply more than 6.5 fl ozs (0.127 lb mefentrifluconazole, 0.042 lb fluxapyroxad) per acre per application.
• DO NOT make more than 2 applications per year.
• DO NOT apply more than 13 fl ozs (0.254 lb mefentrifluconazole, 0.084 lb fluxapyroxad) per acre per year.
• DO NOT apply more than a cumulative total of 0.27 lb ai/acre/year of mefentrifluconazole-containing products corn grown for seed, field corn, and popcorn.
• DO NOT apply more than 0.39 lb mefentrifluconazole-containing products to sweet corn per acre per year.
• DO NOT apply more than a cumulative total of 0.180 lb fluxapyroxad-containing products per acre per year.</td></tr>
</table>

Figure 1: Pesticide label example with a merged cell

- **3.3 Data Structuring and Storage:**
    - **Data Normalization and Schema Mapping:** The extracted information is normalized to ensure consistency (e.g., standardizing units) and mapped to a predefined schema (e.g., with fields like "Crop," "Chemical Name," "Application Rate").
    - **xFM Integration:** The structured data is upserted into the master data database for use in various features and functionalities.
- **4. Business Value and Impact for xFM**
    - **Enhanced Data Quality and Efficiency**: Automating extraction minimizes errors, improves data reliability, reduces time and effort, and enables xFM to scale its database efficiently.

- **Improved User Experience and Competitive Advantage**: Providing accurate and accessible label information enhances user satisfaction, strengthens the value proposition of xFM, and provides a competitive edge.

5. Next Steps:

- **Data Acquisition:** This phase would involve identifying and collecting a comprehensive and representative dataset of pesticide labels from various regions and regulatory bodies (e.g., EPA in the US, similar agencies in other countries). This would also include establishing a process for handling label updates and ensuring data quality and completeness.
- **System Development:** This phase would focus on building the automated extraction pipeline. This would involve selecting and implementing specific tools and libraries for PDF processing (e.g., pdfplumber, PyPDF2), developing the logic for robust table extraction (potentially using a combination of rule-based methods and machine learning models for complex tables), implementing the information extraction techniques (leveraging NLP libraries for semantic analysis, regular expressions for pattern matching, and potentially exploring contextual understanding models), and building the data structuring and storage components (defining a clear and scalable data schema and determining the appropriate database technology for integration with xFM, such as a relational database or a NoSQL solution).
- **Evaluation and Testing:** This critical phase would involve developing a rigorous testing strategy to evaluate the accuracy and reliability of the automated extraction pipeline. This would include creating a manually annotated "ground truth" dataset of pesticide labels, defining key performance indicators (KPIs) such as extraction accuracy, precision, recall, and processing time, and conducting thorough testing and validation to ensure the system meets the required performance standards.
- **Integration into the Master Data Infrastructure:** This final phase would involve developing the necessary APIs and data transfer mechanisms to seamlessly integrate the extracted and structured data into the existing xFM master data infrastructure. This would require close collaboration with the master data team to ensure data compatibility, address any potential conflicts, and establish monitoring and alerting systems for the automated pipeline to ensure its ongoing stability and performance.

6. Conclusion

Automated pesticide label data extraction is a crucial investment for enhancing the capabilities and value of xFM. This proof of concept demonstrates the feasibility of using a multi-stage pipeline and advanced ML techniques to transform unstructured label data into a valuable source of structured knowledge. By implementing this technology, xarvio can improve data quality, increase efficiency, enable new features, and ultimately empower farmers with the information they need for safe and sustainable crop management.